



PARKTOWN BOYS' HIGH SCHOOL
DEPARTMENT OF GEOGRAPHY

GEOGRAPHY

—PRELIM PAPER 2—

27 September 2023

MEMO

Examiner: Mr. D. Grace

Internal Moderator: Mr. A Meintjes

External Moderator: Mr. D. Harvey

TIME: 3 hours

MARKS: 150 Marks

SECTION A – Rural and Urban Settlement

QUESTION 1: Short Questions

- 1.1** Match the term/concepts below to the statements that follow. Write **ONLY** the term/concept next to the question number. Example 1.1.8 Commercial zone.

urban hierarchy; threshold population; urban profile; invasion;
break-of-bulk point; urban sprawl; facadism; site; gentrification; ghetto.

- | | | |
|-------|--|----------|
| 1.1.1 | Minimum number of people needed to make a business profitable. | (1x1)(1) |
| 1.1.2 | Replacement of the original type of land use with another. | (1x1)(1) |
| 1.1.3 | Shows the height of the buildings of a city. | (1x1)(1) |
| 1.1.4 | The shapeless uncontrolled expansion of the city. | (1x1)(1) |
| 1.1.5 | When the old, original front or façade of a building is retained and a new development is built behind the old façade. | (1x1)(1) |
| 1.1.6 | Ranking of urban settlements according to their function. | (1x1)(1) |
| 1.1.7 | The actual place where a settlement grew. | (1x1)(1) |

[7]

- 1.2** Various possible answers are provided for each question. Write down **only the letter** of the correct answer next to the corresponding number.

- | | | |
|-------|--|----------|
| 1.2.1 | The largest city type with significant economic power is known as a... | (1x1)(1) |
| | A megalopolis.
B smart city.
C conurbation.
D metropolis. | |
| 1.2.2 | The smallest rural settlement is a... | (1x1)(1) |
| | A village.
B hamlet.
C town.
D isolated farmstead. | |

- 1.2.3 The largest rural settlement, is a... (1x1)(1)
- A town.
 - B hamlet.
 - C village.
 - D isolated farmstead.
- 1.2.4 The smallest urban settlement is a... (1x1)(1)
- A city.
 - B town.
 - C village.
 - D metropolis.
- 1.2.5 A very small grouping of farmsteads without any services or shops is a... (1x1)(1)
- A minor country town.
 - B village.
 - C hamlet
 - D central place.
- 1.2.6 An example of specialised city is... (1x1)(1)
- A Johannesburg.
 - B the Vatican.
 - C De Aar.
 - D Durban.
- 1.2.7 An example of a gateway/gap town is... (1x1)(1)
- A Worcester.
 - B Pretoria.
 - C Phalaborwa.
 - D Saldanha Bay.
- 1.2.8 The transitional zone is known for... (1x1)(1)
- A having mixed land use.
 - B undergoing change.
 - C being on the edge of the commercial zone.
 - D all of the above.

(8x1)(8)

[8]

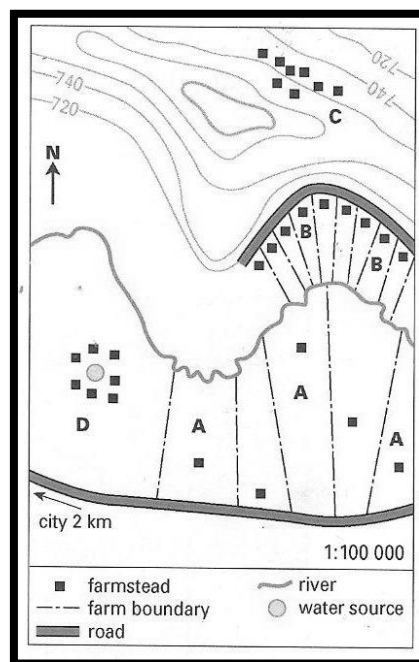
[15]

1	Threshold pop
2	Gentrification
3	Urban Profile
4	Urban sprawl
5	Façadism
6	Urban hierarchy
7	Site

1	A
2	D
3	C
4	B
5	C
6	B
7	A
8	D

QUESTION 2: Classification of Rural Settlements

- 2.1 Refer to the diagram below and answer the questions that follow. Match each of the rural settlements labelled A, B, C and D with one of the following statements:



2.1.1	Irregular and nucleated pattern.	(1x1)(1)
	D	
2.1.2	Irregular and dispersed pattern.	(1x1)(1)
	A	
2.1.3	Nucleated pattern and linear shape.	(1x1)(1)
	C	
2.2	Describe ONE factor that may have influenced the pattern and shape of each settlements labelled A-D.	(4x2)(8)
	[Accept reasonable answers] – MUST BE A DESCRIPTION A – Gradient, access to resources like soil or water, layout of the farm etc. B – Access to the main road C – Defence (on a hill), good view therefore increased property value D – Access to nearby water source	
2.3	Differentiate between the terms site and situation. [Concepts] Site – Actual area a settlement is developed on Situation – Location of a settlement in relation to its surrounding areas	(2x2)(4) [15]

QUESTION 3: Case study

Refer to the article below and answer the questions that follow.

Rural migrants better off in cities.

Moving to a city in search of work seems to pay off for many poor rural South Africans. Data that track changes over time indicate that as many as 385 000

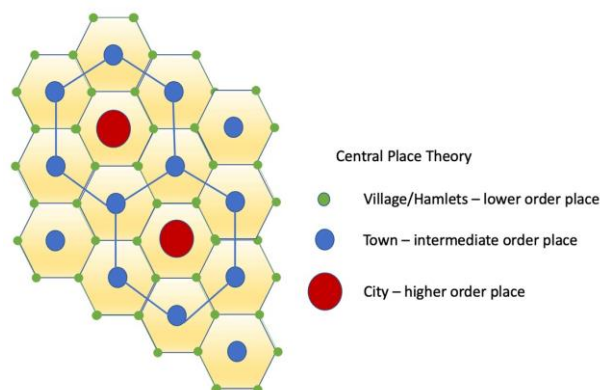
people were lifted from poverty between 2008 and 2014 after moving from rural to urban areas — their poverty levels were halved, together with a fall in unemployment. So government ambivalence about urbanisation should be replaced by a more positive and proactive approach. The fate of people who uproot themselves and migrate to cities is widely discussed. Conventional wisdom says that most end up living in shack settlements, where they struggle to make ends meet because they lack the skills to compete for the paltry jobs available. Their frustrations give rise to protests, land invasions and other forms of antisocial behaviour. Another scenario is that, by sheer determination, they manage to grasp some kind of urban livelihood and gradually claw their way up the job ladder. Historically, there has been a strong relationship between urbanisation and economic development in many parts of the world. Cities function as catalysts for prosperity; they can generate economies of scale and scope, reduced logistics costs and access to shared services, specialised suppliers, thick labour markets and knowledge spillovers. But the benefits of density and proximity are not automatic. Many African cities suffer from negative externalities such as congestion, infrastructure bottlenecks and pollution.

3.1	What do you understand by the sentence “so government ambivalence about urbanisation should be replaced by a more positive and proactive approach.”?	(1x2)(2)
	[Accept reasonable responses] Government should start seeing urbanisation as an opportunity and not a problem.	
3.2	Distinguish between rural depopulation and selective outward migration.	(2x2)(4)
	[Concepts] Rural depop – People are leaving because they have no other choice (accept examples such as climate change, unemployment etc.) Selective Outward Migrat – It is a choice to leave, and often a specific characteristic of person is leaving (men).	
3.3	With reference to the article, discuss the ONE positive and ONE negative consequence of rural-urban migration.	(2x2)(4)
	[Accept reasonable answers] Positive – Potential to become a huge economic boost. Variety of products created. Labour markets climb. Many people are helped at once.	

	Negative – protests, land invasions and other forms of antisocial behaviour arise. Overcrowding and strain on all services	
3.4	Provide and explain ONE cause of rural depopulation.	(1x2)(2)
	[Accept reasonable answers] MUST EXPLAIN Climate change ruining livelihoods. Unemployment and the rise of machines. Promise of a better life. Etc.	
3.5	Name THREE consequences of rural depopulation for the rural settlement.	(3x1)(3)
	[Accept reasonable answers] Less workforce, less security, youth leaves (less care of elderly), less production, less innovation, more corruption, more crime etc.	[15]

QUESTION 4: Central place theory

4. Refer to the diagram below and answer the questions that follow.



4.1	What is an urban hierarchy?	(1x2)(2)
	Ranking of urban settlements according to their function	
4.2	Who developed the idea of central place theory?	(1x1)(1)
	Walter Christaller (be lenient on spelling)	
4.3	Why , according to Central Place theory, do we find more lower order places than higher order places?	(1x2)(2)

	Lower order centres require a far smaller threshold pop and therefore we have more.	
4.4	Name ONE low and ONE higher order function or service suggested by the Central Place theory.	(2x1)(2)
4.5	[Accept reasonable answers] Low – Pizza takeaways, bakeries, butchers, PnP etc. High – Heart surgeon, lawyers, universities etc. In a paragraph of approximately EIGHT lines, discuss the three main ideas that underpin the Central Place theory and provide and explain ONE South African example where this applies.	(4x2)(8) [15]
	[Accept reasonable answers] Main ideas centre around: ▪ People want convenience and the cheapest price ▪ Businesses want to maximise profits ▪ Travelling must be at a minimum Example There are tons of petrol stations around because we will not travel far for petrol. There are tons of McDonalds around because it is a lower-order good and we want convenience We will travel to Cape Town for a specialist operation because it is a high-order service and therefore not common all over	

TOTAL SECTION A: [60]

SECTION B – Economic Geography

QUESTION 5: Short Questions

- 5.1 Provide the correct geographic term or phrase that fits each definition or explanation below. Write only the number and the correct answer.

Column A	Column B
5.1.1 Centripetal force	A: The side view of a city.
5.1.2 Urban profile	B: When old houses close to the city centre are modernised and improved.
5.1.3 Range	C: Changing the existing structure of land tenure; redistribution of land.
5.1.4 GEAR strategy	D: A strategy adopted by the government in 1994 to solve social and economic problems.
5.1.5 RDP	E: Maximum distance that people travel to buy goods.
5.1.6 Gap town	F: Land owned by the community.
5.1.7 Gentrification	G: A gateway into the interior.
5.1.8 PWV complex of Gauteng	H: A plan to promote growth, employment and redistribution.
	I: A force that attracts and keeps people in an urban settlement.
	J: One of the core industrial regions in South Africa.
	K: A force that repels and pushes people out an urban settlement.

[8]

- 5.2 Choose the correct concept/term from those given between brackets to make the statements **TRUE**. Write only the concept/term next to the question number. Example: 2.2.8 Industry.

- 5.2.1 **(Footloose/Ubiquitous)** industries that can be located anywhere. (1x1)(1)
- 5.2.2. An industry that produces less pollution and handles small amounts of raw materials e.g. **(Sasol/bakery)**. (1x1)(1)

- 5.2.3 Industries located close to the consumer are known as **(market/raw material)** orientated industries. (1x1)(1)
- 5.2.4 Industries that produce a lot of pollution and are located near bulk transport networks are known as **(break-of-bulk/heavy)** industries. (1x1)(1)
- 5.2.5 Industries that remain in the same location even though the original reasons for being there are gone are referred to as **(industrial inertia/hinterland)**. (1x1)(1)
- 5.2.6 Mineral **(by-products/beneficiation)** add value to a mineral. (1x1)(1)
- 5.2.7 **(Multiplier effects/Link industries)** are referred to as additional economic developments that stem from the mining industry. (1x1)(1)

[7]

1	L	1	Footloose
2	A	2	Bakery
3	E	3	Market
4	H	4	Heavy
5	D	5	Industrial inertia
6	G	6	Beneficiation
7	B	7	Link industries
8	J		

QUESTION 6: Agriculture

6. Refer to the image below and answer the questions that follow.

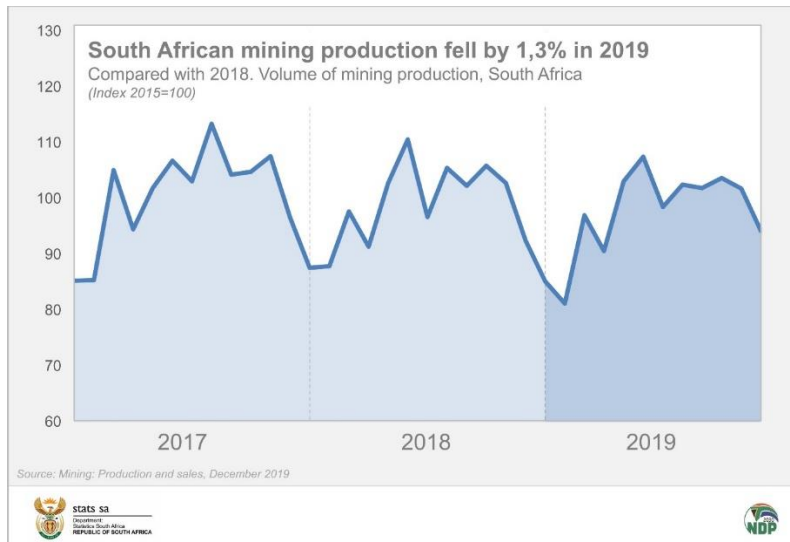


6.1	Name the TWO types of farming practised in South Africa.	(2x1)(2)
	[Any TWO] Large-scale Commercial, small-scale commercial or Subsistence. (Accept commercial)	

6.2	What is the message in the cartoon?	(1x2)(2)
	Africa's farming industry needs to grow/modernise. (Accept "Africa needs young farmers"). Africa's farming industry is not expanding.	
6.3	Provide and explain TWO factors that favour farming in South Africa.	(2x2)(4)
	[Any TWO] MUST BE AN EXPLANATION • We have a wide range of climates • Long growing season • Flat land • Many rivers • We can grow exotic crops here • Climate research • Plant research (GMO crops) • Trade	
6.4	What is food security?	(1x2)(2)
	[Concept] Having sufficient access to food	
6.5	What is the link between poverty and food insecurity?	(1x2)(2)
	[Accept reasonable answers] As food insecurity increases, so does poverty	
6.6	List THREE factors hindering agriculture in South Africa. [Any THREE] • Only 7% of our land is arable • Inconsistent and sometimes insufficient rain • Droughts • Poor soils in most regions • No agriculture subsidies • Bad farming practices • Land ownership • Trade	(3x1)(3) [15]

QUESTION 7: Mining

7. Refer to the figure below showing a graph and answer the questions that follow.



7.1	Which TWO years was South Africa's mining production the highest?	(2x1)(2)
	2017 & 2018	
7.2	How has the discovery and extraction of minerals benefited South Africa? Provide and explain TWO ways.	(2x2)(4)
	[Accept reasonable answers] MUST EXPLAIN Increased exports, improved economy, created jobs, improved investment in SA, improved infrastructure etc.	
7.3	List THREE factors that favour mining in South Africa.	(3x1)(3)
	[Any THREE] • Fluctuations in rand exchange rates • Underground water is dangerous and costly to drain • Labour costs have risen greatly – and of course we love to strike • We are very environmentally conscious – making mining difficult • Cost of transport is high	

	HIV and lung diseases are a major concern	
7.4	What does the above graph suggest about the future of mining in South Africa? [Accept reasonable answers] We are currently in a slump but it is still steady. (Accept that we should probably start looking to invest elsewhere)	(1x2)(2)
7.5	Why should South Africa be concerned about the future of mining? Provide and explain TWO reasons. [Accept reasonable answers] MUST EXPLAIN Mining is finite Machines are replacing people The world markets are changing Environment is suffering etc.	(2x2)(4) [15]

QUESTION 8: Strategies for Industrial Development

8. Refer to the cartoon below and answer the questions that follow.



8.1	In the year 1996 the South African government started an ambitious housing development plan. List TWO reasons	(2x1)(2)
	[Accept reasonable answers] Past injustices needed to be rectified, many people did not have housing, Apartheid meant many black people did not have houses etc.	

8.2	What does BBB-EE stand for?	(1x1)(1)
	Broad-based Black Economic Empowerment	
8.3	Explain the humour in the cartoon.	(2x2)(4)
	[Accept reasonable answers] The politician is telling us that SA business must create jobs SA business can't get going because the country is held back by political incompetence	
8.4	In the context of South Africa's post-Apartheid transformation, elaborate on the fundamental principles and goals of the GEAR policy, focusing on TWO crucial elements. Following this, assess the effectiveness of the BEE policy in promoting black economic empowerment. [Accept reasonable answers] Fundamental Principles [ANY TWO] • Aimed to expand the private sector • Encourage trade • Use better business procedures • Increase output • Increase investment BEE [ANY TWO] Failed the country and destroyed GEAR progress • Only benefitted a few • Enriched very few people and did not distribute money • Created a class of super-rich people that don't have any significant entrepreneurial skills • Makes business consider race before ability • Violates trade policies with foreign mining and industrial companies who have to give up a portion of their profits without benefit • Created the brain drain	(4x2)(8) [15]

TOTAL SECTION B: [60]

SECTION C - Mapwork

RESOURCE MATERIAL

1. 2930CA MERRIVALE and an orthophoto 2930CA 5 MERRIVALE

GENERAL INFORMATION ON MERRIVALE



Merrivale rests not far from the banks of the Howick Dam.

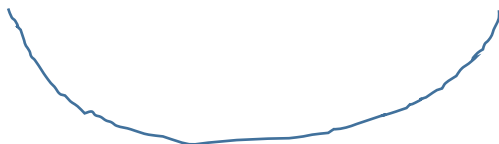
The Main road and the Umgeni River act as dividers. But tacked on to the southern tip of Howick as it is, Merrivale might have its own train station and accompanying old-style character railway houses, but today it merges seamlessly with the larger Howick.

Just alongside Merrivale is the [Sakabula Golf and Country Estate](#), a major draw card to the area for golfers – the only par 73 rated golf course in the province with two very different nine hole courses, set in a rural environment.

There are also a series of falls in the area. [Howick Falls](#), after which the town is named, rush 95m down dolerite cliffs on the Umgeni river, whilst Cascade Falls, Shelter Falls and Karkloof Falls all lie within the vicinity of Howick, and make a good day trip.

QUESTION 9: Calculations

9.1	Provide the grid reference of the map directly South of 2930CA.	(1x2)(2)
	2930CC	

9.2	Calculate the straight line distance between 6 and 7 on the orthophoto in km. Use the formula: <u>$Distance\ on\ a\ map \times scale$</u> $100\ 000$	(1x2)(2)
	9cm = 0.9km	
9.3	Draw a simple cross section between spot height 1422 and spot height 1411 in block E4.	(1x2)(2)
		
9.4	Calculate the magnetic bearing from spot height 1411 to spot height 1422 in block E4. Use the formula: $True\ Bearing + Magnetic\ Declination = Magnetic\ Bearing$ MB = $243^{\circ} + 24^{\circ}42'$ = $267^{\circ} 42'$ (Accept corrected Mag Dec) MB = $243^{\circ} + 25^{\circ}45'$ = $268^{\circ} 45'$	(4x1)(4) [10]

QUESTION 10: Interpretation

10.1	In which province is Merrivale located?	(1x1)(1)
	KZN	
10.2	What type of economic activity is predominate in B5? Provide a reason for your answer.	(2x1)(2)
	Agriculture. It is dominated by cultivated land.	

10.3	What is the dominant street pattern in block C1 of the Topographic map? Name ONE benefit of this pattern.	(2x1)(2)
	Grid iron. Navigation is easier. Faster traffic flow.	
10.4	Account for the location of the sewerage works in block C2.	(2x1)(2)
	[Accept reasonable answers] Health hazard hence far away. Close enough to be useful for the town. Far away for aesthetic purposes.	
10.5	Which area is wealthier between Cedarge (blocks E1 & E2) and Mpophomeni (blocks D2 and D3)? Provide a reason for your answer.	(2x1)(1)
	Cedarge Larger plot sizes, better views, further away from hazards like mines etc.	
10.6	Urbanisation has been increasing in the Merrivale region lately, particularly around block C1. Provide ONE piece of evidence that supports this statement and ONE suggestion to further increase urbanisation in the future. [Accept reasonable answers] Area between settlement and main road has been cleared. There is a recreational space that has already been planned and implemented. Suggestion: Focus on tourism, promote economic development, build industry somewhere.	(2x2)(4) [12]

QUESTION 11: GIS

11.1	Consider the Orthophoto map.	
11.1.1	Is feature 6 on the orthophoto map an example of a point, line or polygon feature?	(1x1)(1)
	Line	

11.1.2	Which format, raster or vector, would be best for showing categories of data such as elevation?	(1x1)(1)
	Raster	
11.1.3	Name the GIS concept that assigns a value or characteristic to each geographical feature.	(1x1)(1)
	Attribute data	
11.1.4	Name ONE layer that would have likely been considered by GIS consultants before developing the powerlines in this area.	(1x1)(1)
	[Accept logical answers] Gradient, consumers, population, rock type etc.	
11.1.5	Explain how you could use GIS in managing the buffer zone around the dams in block C1 and C2 of this map. [Accept reasonable answers] Identify safe areas for development and develop an action plan for government to implement GIS can effectively mark the area for you. Can see all sorts of hazards and help create safe space	(2x2)(4) [8]

TOTAL SECTION C: [30]

GRAND TOTAL:
150 MARKS